

Nature Research Reporting Summary

Manuscript: “Sample-Size Phase Transition Ends the Hard-versus-Soft Clustering Debate”

Corresponding Author: Shuaidong Gao (gsd3247186514@gmail.com)

Statistics

- **Sample sizes:** TCGA cross-cancer comparison: 33 cancer types ($n = 45$ –1218 samples; 10 seeds at $d = 200$, Extended Data Table 1). Within-cancer resampling: 8 cancers, each resampled at $n \in \{50, 100, 200, 400\}$ (5 seeds). Cross-modal replication: MNIST $n \in \{50$ –3000 $\}$ (12 sizes, 3 seeds each); PBMC 10k $n \in \{50$ –10000 $\}$ (15 sizes, 3 seeds each); 20 Newsgroups $n \in \{50$ –10000 $\}$ (15 sizes, 3 seeds each). Methylation: 8 cancer types, $n = 99$ –554 (3 seeds). No statistical method was used to predetermine sample size; all available data were used.
- **Statistical tests:** Spearman rank correlation (two-sided) was the primary test for monotonic relationships between sample size and method advantage. Mann-Whitney U tests compared small-sample ($n < 323$) vs. large-sample ($n \geq 323$) cancer groups. Wilcoxon signed-rank tests evaluated paired CAGate-vs-SSCAGate comparisons. Binomial tests assessed all- Δ -positive claims across the 33-cancer panel. Curve fitting used Levenberg-Marquardt optimization.
- **Data exclusions:** No cancer types were excluded from the cross-cancer comparison. The number of top variable genes was capped at $d = \min(200, n - 10)$ per cancer to ensure identifiability ($d < n$); for all cancers with $n \geq 210$ this yields $d = 200$. For methylation gene aggregation, genes with < 3 CpG probes were excluded (standard practice).
- **Replication:** The phase transition was replicated in: (i) 33 independent TCGA cancer types, (ii) 8 within-cancer resampling experiments, (iii) PBMC single-cell RNA-seq datasets (3k, 6k, 10k), (iv) MNIST image data, (v) 20 Newsgroups text data, (vi) 8 methylation datasets. All replication attempts confirmed the phase transition.
- **Randomization:** For all subsampling experiments, random seeds (42 for MNIST, 42 for PBMC 10k, 43 for 20 Newsgroups) were used to select cell/image/document subsets. For CAGate/SSCAGate runs: 10 seeds per cancer for the $d = 200$ cross-cancer comparison (Extended Data Table 1), 5 seeds for within-cancer resampling (Figure 1b), and 3 seeds for cross-modal replication. K-means initialization was randomized.
- **Blinding:** Not applicable. All analyses were reproducible given fixed random seeds.

Experimental Design

- **Study design:** This is a computational study with three independent experimental paradigms: (i) cross-cancer comparison of 33 independent datasets, (ii) within-cancer controlled resampling, and (iii) cross-modal replication on completely unrelated data types. No treatment groups or interventions were involved.
- **Controls:** Standard NOTEARS (without gating) served as the baseline in all experiments. Synthetic data (Erdős-Rényi DAGs) were used as a negative control. Random cluster assignment was used as an ablation control.
- **Data sources:** All data are publicly available. TCGA gene expression: GDC Data Portal and UCSC Xena. Single-cell RNA-seq: 10x Genomics and Scanpy. MNIST: Yann LeCun’s website. 20 Newsgroups: public NLP corpus. DNA methylation: Firehose Broad GDAC.

Software and Algorithms

- **Programming languages:** Python 3.12
- **Key packages:** PyTorch 2.11, scikit-learn 1.6, Scanpy 1.10, NumPy 1.26, SciPy 1.11
- **Custom code:** CDSM framework (v3.0.0), available at <https://github.com/gsd3247186514-cell/SSCAGate-Nature> (MIT license)
- **Algorithms:** CAGate (two-stage cluster-aware gradient gating), SSCAGate (self-supervised causal gating with end-to-end cluster optimization), NOTEARS (Zheng et al., 2018)
- **Hardware:** AMD Ryzen 9 8945HX (16-core CPU), NVIDIA RTX 5060 Laptop GPU (8GB), 16GB DDR5 RAM, Windows 11

Data and Code Availability

- **Data availability statement:** Included in the main manuscript. All datasets are publicly accessible. Full replication package at <https://github.com/gsd3247186514-cell/SSCAGate-Nature>.
- **Code availability statement:** Included in the main manuscript. Full source code at <https://github.com/gsd3247186514-cell/SSCAGate-Nature>.

Competing Interests

A Chinese patent application covering the Cluster-Aware Gating mechanism has been filed (May 2026). The author declares no other competing financial or non-financial interests.